Renfred Alonge

Full-Stack Software Engineer · Founder & CTO, Penciil
Mississauga, ON · +1 (647) 457-6660 · alongerenfred@gmail.com · renfred.dev · linkedin.com/in/renfred-ai · github.com/renfrrd-ai

Summary

Full-stack engineer with four years of shipping production software before university, including a learning platform used by 700+ students. Founder of Penciil, an assessment platform for schools whose MVP I built solo.

Education

University of Toronto Mississauga Sep. 2025 – Jun. 2029 (expected)
Honours B.Sc., Computer Science and Mathematics *Mississauga, ON*

- GPA 4.0 / 4.0. Lester B. Pearson International Scholar (2025 cohort); Dean's List (Year 1).
- Incoming summer research with Prof. Lisa Zhang on threshold concepts in machine learning: the ideas that, once grasped, change how students understand the field.

Experience

Founder & Chief Technology Officer 2026 – Present
Penciil · Assessment Platform for Schools *React.js, Node.js, PostgreSQL, Redis, Docker*

- Built the end-to-end assessment workflow schools run on: a lecturer creates an exam, scripts are scanned in and graded, and the results flow straight into course analytics and feedback for each student, all in one system instead of separate tools.
- Designed it around the people who run assessment. Lecturers, teaching assistants, heads of department and faculty administrators each get their own role, permissions and view.
- Designed the PostgreSQL schema and REST APIs for authentication, courses, exam delivery and grading as one data model, so analytics and feedback read from the same records graders write to instead of separate exports.
- Moved the heavy work (grading runs, analytics) onto asynchronous jobs with Redis-backed caching, so it runs in the background without slowing the app. Containerized each service with Docker, with separate development and production setups.
- Expanding Penciil into a suite of course tools modelled on what U of T runs (MarkUs, Piazza, MathMatize). First up is Folio (folio.penciil.com), a rebranded fork of MarkUs, U of T's open-source grading tool. It tracks upstream and changes only the UI, so security and bug fixes keep flowing in without maintaining a diverging codebase.
- Run the business side too. I wrote the PRD, market analysis, ideal customer profile, buyer personas and go-to-market plan; I am targeting open universities and secondary schools in Nigeria first, before slower-moving traditional universities. I also built the brand, website and domain, and filed the trademark.

Co-Founder 2026 – Present
Clueline · AI Error Monitoring (In Development) *Next.js, React.js, Node.js, PostgreSQL, LLMs*

- Building an error-monitoring tool with two audiences: end users get a plain-English explanation of what went wrong; developers get stack traces, session context, and an LLM-generated root cause and suggested fix.
- Split the SDK into a core package plus framework-specific packages (`@clueline/core`, then React and Next.js) instead of one universal package. The SDKs live in a monorepo, separate from the dashboard and ingestion API.
- The fix suggestions read the developer's connected GitHub repository, so they point at the actual code instead of guessing from the stack trace alone. Repo connection is available on every tier, not locked behind a paid plan.
- Wrote the PRD, architecture doc and phased build plan; positioned the product against Sentry by serving end users as well as developers.

Full-Stack Software Engineer Nov. 2021 – Jul. 2025
SMATHS-Kids Initiative · Educational Platform *Ogun State, Nigeria*

- Built and maintained smathskids.com, the production learning platform for 700+ active students, as its main engineer: student accounts, course content, assessments, and the admin dashboards staff used for grading and tracking progress.
- Designed the REST backend behind grading, analytics and student tracking, so every dashboard read from the same student records rather than each workflow keeping its own.
- Automated the reports staff had been assembling by hand. Data pipelines now generate student and class reports directly from platform data, cutting manual administrative work by about 40%.

Software Engineer & Automation Developer Jan. 2023 – Jul. 2025
Apt Scholars Universal College *Ogun State, Nigeria*

- Built Python tools that generate quizzes, grade them, and turn the results into per-student performance reports, replacing the manual process teachers repeated every term.
- Wrote Pandas and Excel-automation pipelines that take score sheets from raw marks to finished reports, cutting grading turnaround by about 50%. Packaged them as reusable scripts other teachers could run without help.
- Also taught STEM: named 2nd Best Teacher (Cowbellpedia STEM 2025) and coached students to national competition finals.

Teaching & Leadership

- Designed and presented the workshop *Why the Best Path Sometimes Goes Backwards* at the Pearson Conference 2026 (U of T, Bahen Centre), teaching local optima through "The Hiker in the Fog," an interactive game I designed where participants must move away from the goal to eventually reach it.
- Run exam-prep sessions for classmates before midterms and finals; reviewed student workshop applications on the conference's review panel.
- Member, Pearson Scholars Social Planning Committee: organize campus-wide community events.

Technical Skills

Languages: Python, JavaScript, Java, SQL, C · **Web:** React.js, Next.js, Node.js, Express.js, FastAPI, Tailwind CSS
Data & Infra: PostgreSQL, Redis, Docker, Git, Linux, REST API design · **AI/ML:** LLM integration, Scikit-learn, Pandas, NumPy